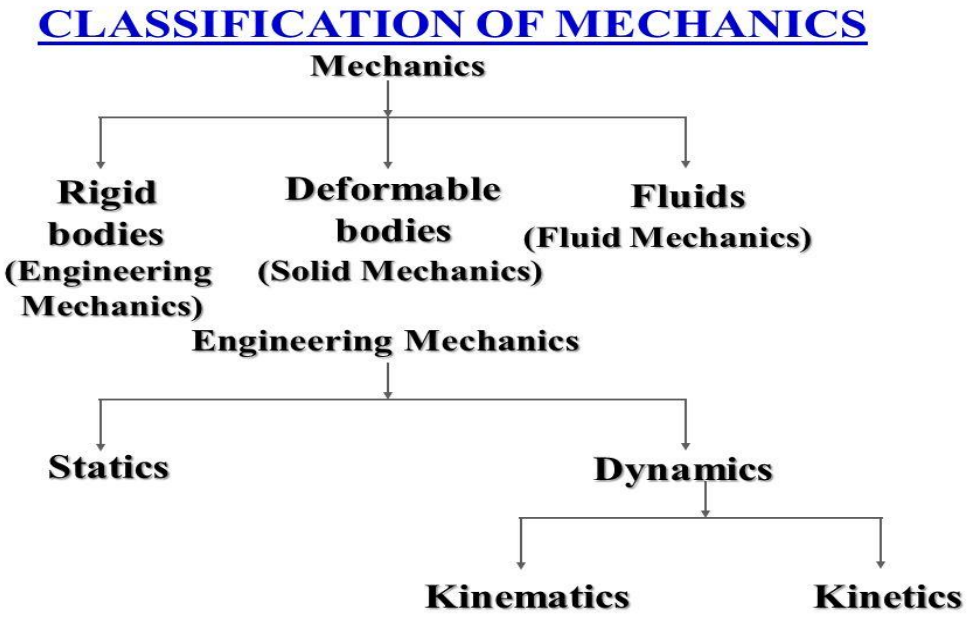
# Module 1

## Introduction to Civil Engineering

**Civil Engineering:** **Civil engineering** is a professional engineering discipline that deals with the design, construction, and maintenance of the physical and naturally built environment, including public works such as roads, bridges, canals, dams, airports, sewage systems, pipelines, structural components of buildings, and railways.

## Fundamentals

**Mechanics:** Mechanics is a branch of the physical sciences that is concerned with the state of rest or motion of bodies subjected to the action of forces. Engineering mechanics is the application of mechanics to the solution of engineering problems or branch of mechanics which deals with the action of forces on rigid bodies.:



Mechanics broadly classified into following

- ✓ Mechanics of Rigid bodies
- ✓ Mechanics of deformable bodies
- ✓ Mechanics of fluids

**Mechanics of Rigid bodies**

It is the branch of science which deals with the study of bodies that do not undergo any deformation under the application of forces or a rigid body is a system of particles that don't change their relative positions, even when the body moves. It can be further classified into two types

- ✓ Statics
- ✓ Dynamics.

**Statics:** Branch of mechanics which deals with the study of behaviour of bodies in state of rest.

**Dynamics**: Branch of mechanics which deals with the study of behaviour of bodies in state of motion. It can be further classified into two types

- ✓ Kinematics
- ✓ Kinetics

Kinematics: The forces causing the motion of the body are not considered.

Kinetics: The forces causing the motion of the body are mainly considered.

## Technical Terms used in Engg Mechanics or Idealization in Mechanics

**1. Particle:** A particle may be defined as an object which has only mass and no shape and dimensions. Theoretically, such a body cannot exist; while dealing with the problems involving distances which are considerably larger compared to the size of the body, the size of the body is generally neglected without sacrificing the accuracy. Example are as follows:

Eg: A ship in a mid-sea is a particle in the study of relative motion from the control in a part.

Eg: In the study of the movement of earth in celestial space, earth is treated as a particle.

2. **Continuum:** A body consists of several particles. Again, each particle can be sub-divided into molecules, atoms and electrons. It is not feasible to solve any engineering problem by treating a body as a composite of such discrete particles. Hence the body is assumed to consist of a continuous distribution of matter which will not separate even when various forces considered are acting simultaneously. In other words, we say the body is treated as a continuum.

3. **Rigid Body:** A body is said to be rigid when the relative movements between its particles are negligible or positions of the constituent particles do not change under the action of external forces Eg: When the system of forces F1, F2 and F3 are applied to the body and if body is treated as a rigid body, the relative position of A to B is the same as A' and B', i.e., AB = A'B'

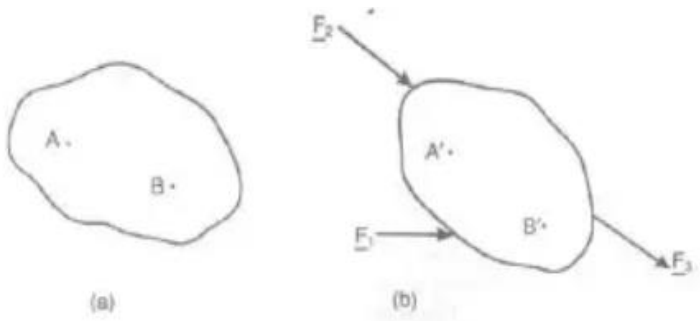



Fig.1: Rigid Body

**4. Deformable body:** A deformable body is one in which the positions of constituent particles change under the application of external forces. For the below figure, the relative positions of 1 and 2 is not same under the action of forces.

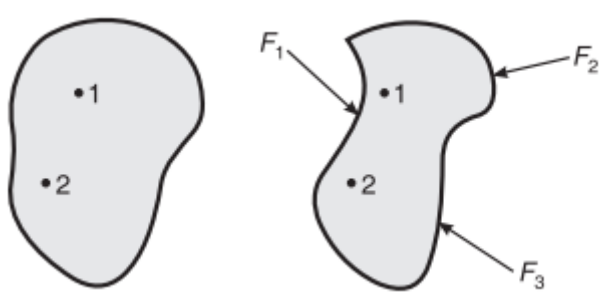



**Fig.2: Deformable body**

**Mass (m):** The total amount of matter present in a body is known as its mass. The unit of mass is the kilogram, abbreviated kg.

**Weight:** A body is attracted towards the earth due to gravitation. This causes acceleration directed towards the centre of earth. It is called acceleration due to gravity and is denoted by g. The resulting force is equal to the weight of body.

Weight = mass × acceleration due to gravity W = mg, in Newton, where g = 9.81 m/s2

**Scalar quantity**: A physical quantity which has only magnitude is called scalar quantity. For example, time, mass, density, volume, distance, and so forth.

**Vector quantity:** A physical quantity which has a direction in addition to magnitude is known as vector quantity. For example, force, displacement, velocity, acceleration, and so forth.

## Force

It is the external agency which tends to change the state of a body or a particle. When a force is applied to a body which is at rest, the body may remain in the state of rest or it may move with some velocity. The SI unit of force is Newton.

### Characteristics of force

A force can be identified by its four characteristics:

**(i) Magnitude:** It can be denoted as 10kgf or 100N. Magnitude of force is the number that represents strength of force.

**(ii) Direction:** The direction of a force can be represented by an arrowhead.

**(iii) Line of action:** It is the line along which the force acts.

**(iv) Point of application:** It is the point at which the force acts.

**Effects of forces**

**External effects**

The forces exerted on bodies of a given system by bodies situated outside the system are called external forces. External forces on a rigid body are due to causes that are external to the body. They can cause the body to move or remain at rest as a whole.

For example: Force of gravity, external force applied on an object.

**Internal effects**

The forces of interaction between bodies composing a system are called internal forces. Internal forces develop in any body that keeps all the particles of a body together.

For example: Internal Tension or compression in a bar, bending moment in a beam.

## Force system

If two or more forces are acting on a body or a particle, then it is said to be a force system, such as that shown in below Figure.

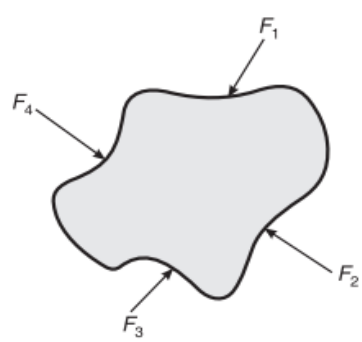



Fig.3: Force System

**Classification of force system**

1. Coplanar force system
2. Non-coplanar force system

3. Collinear force system.

1. **Coplanar force system** : If two or more forces are acting in a single plane, then it is said to be a coplanar force system or it is a force system in which all the forces lie in a single plane.

The types of coplanar force system are:

**a) Coplanar concurrent force system:** If two or more forces are acting in a single plane and their lines of action pass through a single point, then it is said to be a coplanar concurrent force system. Ex: Forces in a rope and pulley arrangement.

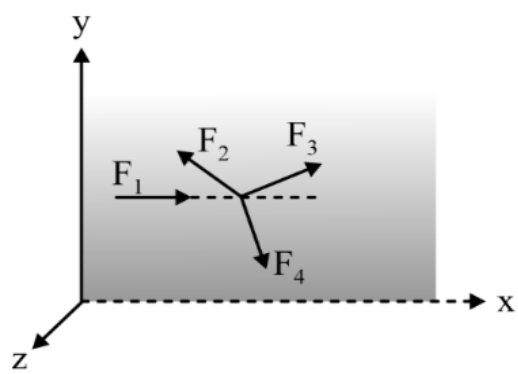



Fig.4: Coplanar concurrent force system

**b) Coplanar non-concurrent force system**: If two or more forces are acting in a single plane and their lines of action do not meet at a common point, then the forces constitute a coplanar non-concurrent force system. Ex: Forces on a ladder and reactions from floor and wall

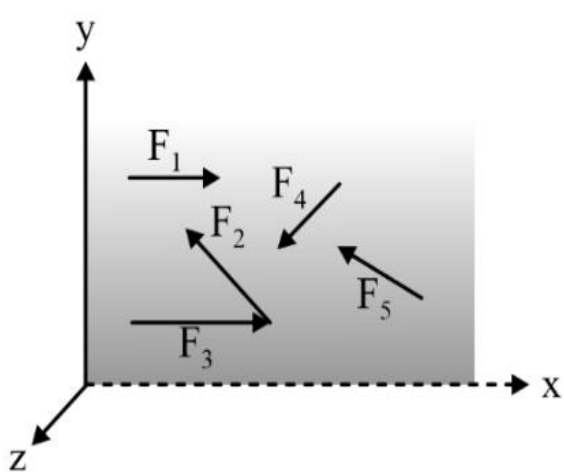



Fig.5: Coplanar non-concurrent force system

**c) Coplanar parallel force system:** If two or more forces are acting in a single plane with their lines of action parallel to one another, then it is said to be a coplanar parallel force system.

The coplanar parallel force system is of two types:

**(i) Like parallel force system:** All the forces act parallel to one another and are in the same direction in the same plane.

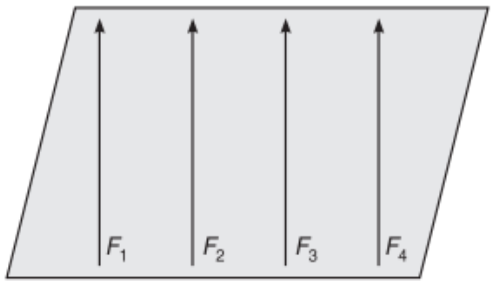



Fig.6: Like parallel force system

ii) **Unlike parallel force system:** The forces act parallel to one another, but some of the forces have their line of action in opposite directions.

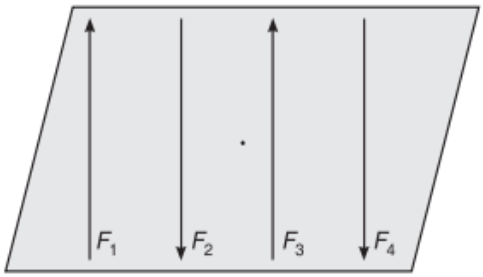



Fig.7: Unlike parallel force system

**Non-coplanar force system**

If two or more forces are acting in different planes, the forces constitute a non-coplanar force system.

a) **Non-coplanar concurrent force system:** If a system has two or more forces acting on different planes but pass through the same point, then it is said to be a non-coplanar concurrent force system. Ex: Forces acting on a tripod when camera is mounted.

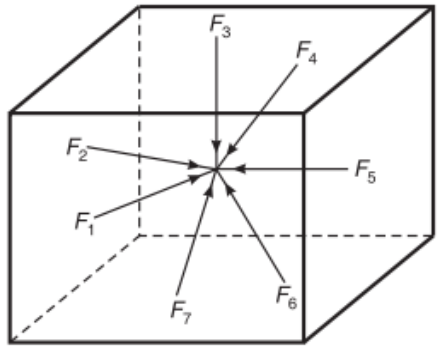



Fig.8: Non Coplanar concurrent force system

**b) Non-coplanar non-concurrent force system:** If two or more forces are acting on different planes but do not pass through the same point, they constitute a non-coplanar non-concurrent force system. Ex: Forces acting on a building frame.

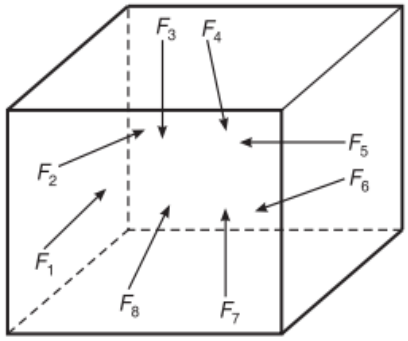



Fig.9: Non Coplanar non- concurrent force system

**c) Non-coplanar parallel force system:** If two or more forces are acting in different planes and are parallel to one another, the system is said to be a non-coplanar parallel force system. Ex: Forces acting and reaction at the points of bench with floor

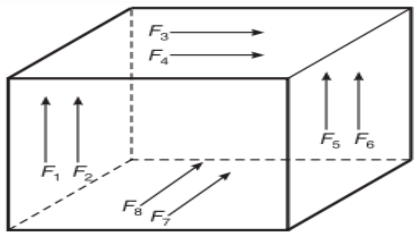



Fig.10: Non Coplanar parallel force system

**Collinear force system:** If the lines of action of two or more forces coincide with one another, it is called a collinear force system.

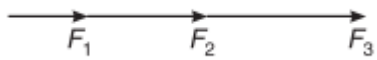



Fig.11: Collinear force system

**Non-collinear force system:** If the lines of action of the forces do not coincide with one another, it is called a non-collinear force system.

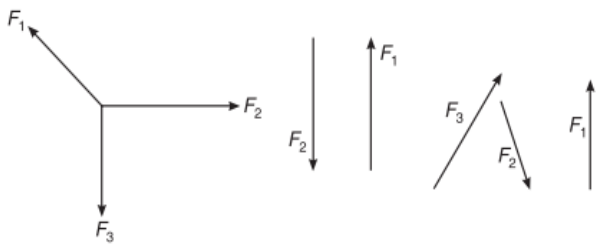



Fig.12: Non-Collinear force system

## Principle of Transmissibility of Forces

This principle states that "A force acting on a rigid body can be transmitted from one point to another point along the same line of action such that the effect produced by the force on a body remains unchanged"

The principle can be applied when a force acts anywhere along the same line of action on a rigid body, as long as the force's magnitude and direction remain the same.

Let us consider a rigid body subjected to a force of *F* at point *O* as shown in below figure, According to the principle of transmissibility, the force *F* can be transmitted to a new point *O*' along the same line of action such that the net effect remains unchanged.

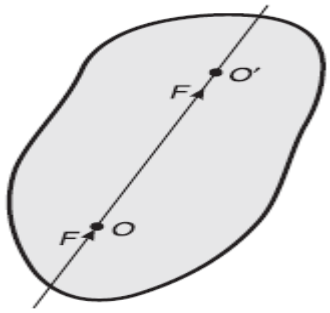



Fig.13: Transmissibility of force F from point O to O'

## Principle of Superposition of Forces

This principle states that the combined effect of force system acting on a particle or a rigid body is the sum of effects of individual forces.

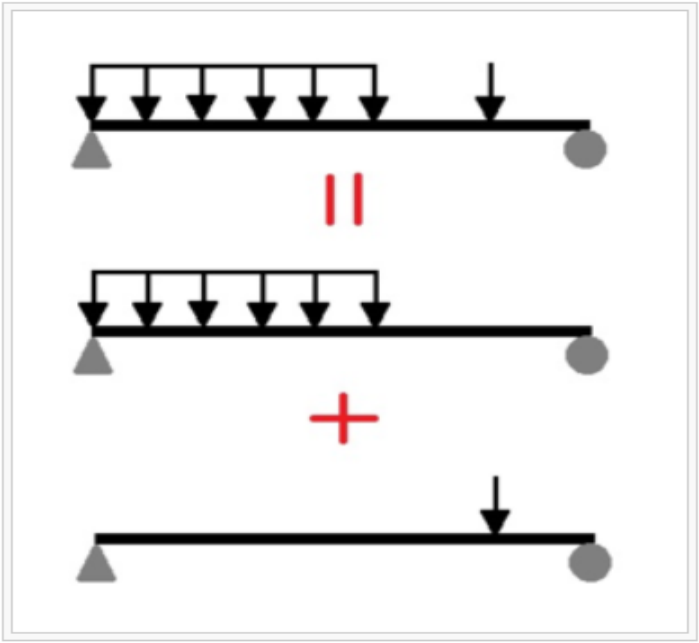


Fig.14: Principle of Superposition

A given beam and its loadings can be split into simpler beams and loading. As shown in above figure , the beam with the distributed load and the point load can be split into two beams. One beam with the distributed load and the other with the point load. According to the principle of

superposition of forces, the combined effect produced by the beam with distributed load and point load will be the same as the effect produced by the individual loadings.

## Action and Reaction forces

Action and reaction forces are two forces that act on two interacting objects, and are described by Newton's third law of motion. The law states that when one object applies a force (action) on another object, the second object applies an equal and opposite force (reaction) back on the first.

An action force is a force that is applied to an object. A reaction force is a consequence of an action force which is opposite in direction.

**Examples:**

- **Pushing Against a Wall:**
  - When you push against a wall, your hand exerts a force (action) on the wall. The wall pushes back with an equal force (reaction) in the opposite direction.
- **Rocket Propulsion:**
  - The rocket engine expels gases downward (action), and the rocket moves upward (reaction) with an equal and opposite force.
- **Walking:**
  - When you walk, your foot pushes backward on the ground (action), and the ground pushes your foot forward (reaction), allowing you to move forward.

## Resolution of a Force

The process of splitting of a force into its two rectangular components (horizontal and vertical) is known as resolution of the force.

**Method of Resolution**

*F* is the force which makes an angle θ with the horizontal axis, and has been resolved into two perpendicular components, namely *Fx* and *Fy*, along the *x*-axis and *y*-axis respectively.

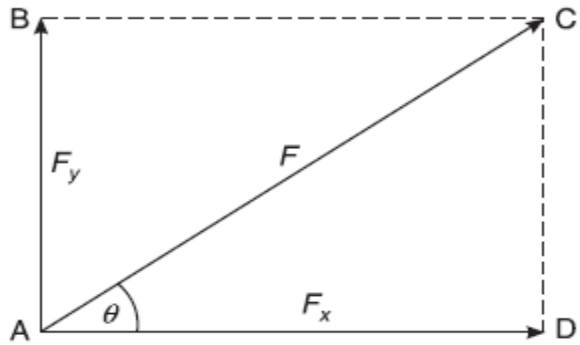



Fig.15: Resolution of a force.

In CAD,

$$\cos\theta = \frac{F_x}{F} \quad \Rightarrow \quad F_x = F\cos\theta$$

$$\sin\theta = \frac{F_y}{F} \quad \Rightarrow \quad F_y = F\sin\theta$$

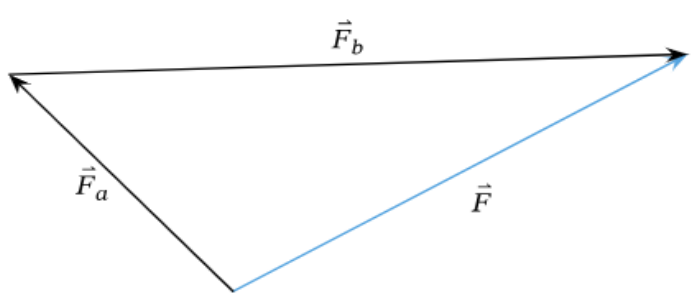
If, on the other hand, $\theta$ is the angle made by the force $F$ with the vertical axis, then

$$F_y = F\cos\theta\,; \quad F_x = F\sin\theta$$

**Note:** If the force $F$ makes an angle of $\theta$ with the horizontal, the horizontal component of the force is $F\cos\theta$.

Where $F_x$ and $F_y$ are the perpendicular components of the force F

We can consider a force as consisting of two non perpendicular components. The following figure shows a force $\vec{F}$ and two non perpendicular components $\vec{F}_a$ and $\vec{F}_b$,where $\vec{F}=\vec{F}_a+\vec{F}_b$.



## Composition of Forces

It is the process of combining a number of forces into a single force such that the net effect produced by the single force is equal to the algebraic sum of the effects produced by the individual forces. The single force in this case is called the **resultant force** which produces the same effect on the body as that produced by the individual forces acting together.

## Method of Resolution

Let $F_1$,$F_2$, $F_3$ and $F_4$ be the forces acting on the body as shown below, $F_1$ makes an angle $\theta_1$ with the horizontal and $F_3$ makes an angle $\theta_2$ with the vertical.

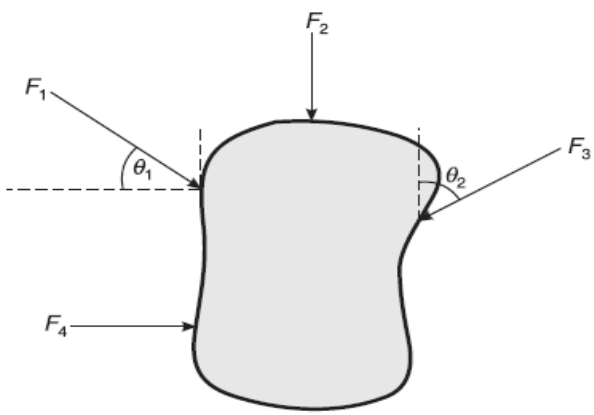



Fig.16: Resolution of forces

From the above fig,

$$\Sigma F_x = \text{algebraic sum of the components of the forces along the } x\text{-axis}$$

$$\Sigma F_x = F_4 + F_1 \cos\theta_1 - F_3 \sin\theta_2$$

$$\Sigma F_y = \text{algebraic sum of the components of the forces along the } y\text{-axis}$$

$$\Sigma F_y = -F_2 - F_1 \sin\theta_1 - F_3 \cos\theta_2$$

∴ The magnitude of the resultant,

$$R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

and the direction of the resultant,

$$\theta = \tan^{-1}\left(\frac{\Sigma F_y}{\Sigma F_x}\right)$$

**Triangle law:** If two forces acting simultaneously on a particle can be represented both in magnitude and direction by the two sides of a triangle taken in order, then the magnitude and direction of the resultant can be represented by the third side of a triangle, taken in opposite order.

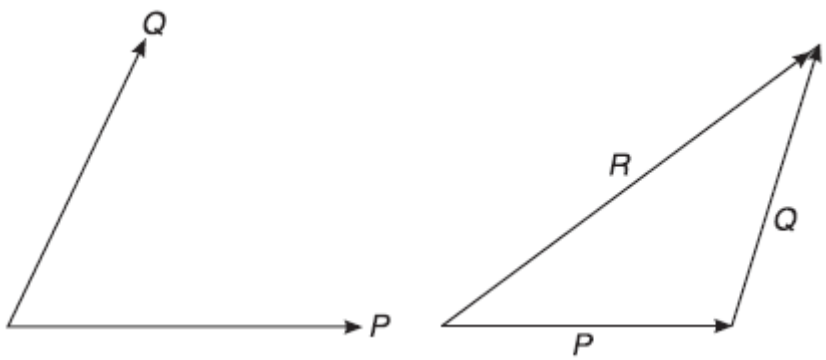



Fig.16: Triangle law of forces

## Resultant of Coplanar Concurrent force system

In a coplanar concurrent force system, we can calculate the magnitude and direction of the resultant as shown below,

The magnitude of resultant,

$$R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

Direction of resultant,

$$\theta = \tan^{-1}\left(\frac{\Sigma F_y}{\Sigma F_x}\right)$$

The steps to solve problems in the coplanar concurrent force system are, therefore, as follows:

1. Calculate the algebraic sum of all the forces acting in the $x$-direction (i.e. $\Sigma Fx$) and also in the $y$-direction (i.e. $\Sigma Fy$).
2. Determine the magnitude of the resultant, R
3. Determine the direction of the resultant, $\theta$

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